

Good Data, Bad Data

Project: Helpful Classroom Sorter

1. Recall and choose your starting point

Try the recall before opening notes. Then use a different colour or a labelled second line to correct your explanation.

Recommendations

Without opening the old lesson, explain one key idea from Recommendations. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

Confidence Is Not Correctness

Without opening the old lesson, explain one key idea from Confidence Is Not Correctness. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

Save New Examples for Testing

Without opening the old lesson, explain one key idea from Save New Examples for Testing. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

2. Practise with fading help

Use W2: Write a traceable correction for Data quality and documented repair. Record the answer once; the lesson and workbook refer to the same attempt.

Given inputs · Procedure I chose · Middle steps · Result and check

Given inputs: _____

Procedure I chose: _____

Middle steps: _____

Result and check: _____

The step I completed without help, and the hint I needed

3. Individual reasoning check

Use the worked case for Data quality and documented repair. Proposed explanation: “Every valid value is true”. Decide whether this explanation is supported. Point to the step or supplied fact that decides; explain your repair if needed.

My explanation before partner discussion

4. Independent transfer case

Independent case: Audit an independent paper-plane log

Each flight ID must be unique; distance is a whole number from 0 through 20 cm. Raw rows: F1 8 cm; F2 120 cm; F3 blank; F4 0 cm; another F1 8 cm. Source S1 confirms one F1 flight. S2 says F2 travelled 12 cm. S3 is blank. S4 confirms zero. Preserve the raw rows, describe supported repairs and give the number of distinct flights. Report how many assessed flights travelled at least 10 cm, including missing records separately.

My independent explanation, trace and limits

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

5. Your learning target

A table lists leaf lengths 8 cm, 90 mm, blank and 700 cm. Propose a documented cleaning action for each row. Which values can you safely change immediately?

6. Solve the given case

Ordinary pencil lengths are recorded as 12 cm, 14 cm, blank and 130 cm.

What needs checking? May 130 silently become 13?

Show your trace, calculation or evidence

Write or draw your reasoning and evidence.

7. Check a changed case

A table labels a cat photo dog and lists the same photo three times. Identify both data problems and one sensible repair for each.

My independent answer and reason

Write or draw your reasoning and evidence.

Week 21 Test and reflect

8. Design a revealing test

Create a changed input, boundary case or missing-information case. Write the expected answer or safe response before testing.

Test record · My evidence

Test record: Changed input or condition

My evidence: _____

Test record: Expected answer or fallback

My evidence: _____

Test record: Observed result

My evidence: _____

Test record: Why the result occurred

My evidence: _____

9. Project checkpoint

Audit the sorter's example cards and add a reason for including each field and condition.

The project change and evidence I will keep

Write or draw your reasoning and evidence.

10. Self check

I can explain the mechanism. I can show evidence. I can handle a changed case. I can name a limit. Mark each as not yet, with help or independently, and attach supporting work.

Teacher review _____ Next step _____

11. Transfer practice from the original programme

A table lists leaf lengths 8 cm, 90 mm, blank and 700 cm. Propose a documented cleaning action for each row. Which values can you safely change immediately?

My independently reasoned answer

Record support used: none, hint or teacher explanation.

12. W1 Check a schema without guessing

Use the RampLog schema. State the issue or valid status for height 60, travel blank and travel 0. Give the range allowed for each numeric field and explain why a blank and zero require different treatment.

My reasoning and evidence

13. W2 Write a traceable correction

C2 says T2 height is 6 cm and travel is 5 cm; the export says height 60 cm. Write a change-log entry with ID, field, old and new values with units, evidence, reason, reviewer and date. Use a fictional reviewer and date. Explain where the raw export is kept.

My reasoning and evidence

14. W3 Distinguish duplicate rows from similar trials

The export repeats T1 with the same source C1, which confirms one trial. A different trial T5 also has height 4 and travel 3, with its own source C5 confirming a separate trial. Explain which rows count once and why T5 must remain separate.

My reasoning and evidence

15. W4 Audit an independent paper-plane log

Each flight ID must be unique; distance is a whole number from 0 through 20 cm. Raw rows: F1 8 cm; F2 120 cm; F3 blank; F4 0 cm; another F1 8 cm. Source S1 confirms one F1 flight. S2 says F2 travelled 12 cm. S3 is blank. S4 confirms zero. Preserve the raw rows, describe supported repairs and give the number of distinct flights. Report how many assessed flights travelled at least 10 cm, including missing records separately.

My reasoning and evidence

16. W5 Handle conflicting and converted evidence

Two signed cards disagree about the same F3 flight: one says 9 cm and the other 11 cm. Neither is established as the correction. What can you report for the at-least-10-cm question? Separately, a source for a new F5 flight explicitly records 100 mm. Convert it to centimetres and state the conversion evidence to retain.

My reasoning and evidence

17. Feedback, return check and learning record

Evidence · My record

Evidence: Worked alone / hint / model

My record: _____

Evidence: First step I repaired and why

My record: _____

Evidence: New case checked later; date and result

My record: _____

Evidence: What I can now explain without notes

My record: _____

Evidence: One remaining question

My record: _____

Revisit this idea with your teacher after a delay. Familiarity is different from explaining and using it on a changed case.

18. Pictorial case practice: Good Data, Bad Data

A fictional toy-car distance table needs cleaning before use.

Contract: unique run ID; whole distance 0–30 cm. Raw rows: A=0 cm, B=200 mm, C=blank, another A=0 cm.

Source log confirms one A run and B's original measurement of 200 mm; C has no measurement.

Good Data, Bad Data

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Validate rows

A's zero is valid; B has wrong unit; C is missing; A is duplicated.



2. Evidence repair

200 mm / 10 = 20 cm; source log supports retaining one A.



3. Keep unknown

C stays unresolved; do not replace it with 0.



4. Clean record

Usable unique rows: A=0 cm, B=20 cm; retain a separate change log.

CHANGE ONE THING

The source log gives B only as bare 200 with no unit.

Explain the changed result using the matching lesson.

Which repairs are evidence-backed in the original case, and which row must remain unknown?

My picture-based explanation and changed-case prediction

Support used: alone / hint / worked example. This record is practice evidence.

Project connection: Keep raw sorter inputs, validated records and an evidence-linked repair log as different artifacts.

19. Project portfolio milestone

Project: Helpful Classroom Sorter

Current milestone: Evaluate and revise. Compare with your starting method, explain errors and record a justified revision.

Keep raw sorter inputs, validated records and an evidence-linked repair log as different artifacts.

Evidence to keep · My record

Evidence to keep: This week’s input and deciding step

My record: _____

Evidence to keep: Prediction and actual result (or labelled paper trace)

My record: _____

Evidence to keep: One change since the previous checkpoint and why

My record: _____

Evidence to keep: One error or limit and the next test

My record: _____

Explain what this week adds to your project. Reuse last week’s record rather than starting again.

20. Supported practice and learning evidence

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

A fictional toy-car distance table needs cleaning before use.

Practice 1: explain the deciding evidence

Which labelled step matches this detail? Usable unique rows: A=0 cm, B=20 cm; retain a separate change log.

1. Clean record
2. Evidence repair
3. Keep unknown
4. Validate rows

My selected conclusion, deciding fact and explanation

Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

After your attempt, compare with the supported explanation: Clean record Usable unique rows: A=0 cm, B=20 cm; retain a separate change log. Cleaning is not permission to make inconvenient values plausible. The source log supports a unit conversion and removal of a duplicate copy. Zero means a measured zero distance; blank means no supplied measurement. Keep the raw table and explain every change so someone can reconstruct what happened.

Practice 2: explain the deciding evidence

Changed-case practice: The source log gives B only as bare 200 with no unit. Which conclusion is supported by the supplied facts?

1. B remains unresolved rather than becoming 20 cm.
2. We can decide the result without examining the changed input or the procedure.
3. This one example guarantees correct results for every future input.

My selected conclusion, deciding fact and explanation

Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

After your attempt, compare with the supported explanation: B remains unresolved rather than becoming 20 cm. Without a confirmed unit, division by 10 would be an unsupported assumption.

Repair a missing building block

Review Recommendations (ai-1/20). Goal: Apply eligibility requirements before ranking candidates by preferences. Short practice: Why does the recommendation change without changing anyone's preferences?. Return to this week and explain what you changed.

Review Confidence Is Not Correctness (ai-1/18). Goal: Distinguish a confidence claim from evidence that a prediction was correct. Short practice: Was the unknown label guaranteed correct because its confidence was 80%?. Return to this week and explain what you changed.

Review Save New Examples for Testing (ai-1/15). Goal: Separate examples used to choose a model from examples reserved for its final evaluation. Short practice: Does 4/4 on the replay replace the original 3/4 result? Explain.. Return to this week and explain what you changed.

Keep your completed work

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run evidence. Open the HTML copy to read or print it. Drafts stay in this browser when storage is available; clear drafts on a shared device.

What I repaired, the evidence I kept and the next test
